

Infinite Zoom Is a Property of Representation, Not of Fluids

Ray David Crowell

Cognitive Systems Incorporated

Ray@CogSysAi.com

<https://github.com/RayFromBoston/Public-Research>

For the 3D incompressible Navier–Stokes field, either every nice initial velocity stays smooth for all time, or there is a finite-time blow-up. Prove one. Fefferman, C. L., Clay Mathematics Institute, 2000.

The answer to Fefferman’s dichotomy is neither box, because his equation does not correctly describe fluid at all scales. And here is a different, correct equation that does.

Clay’s formal PDE cannot solve this: as soon as you switch the fluid molecule, any proof you created is disproven. Different fluid molecules have different properties, and those properties change whether the fluid stays smooth or breaks. Period. The right tool is the molecular floor plus Lemma G, provided in this paper. The answer is not a single yes-or-no for all fluids. Fluids themselves do not infinitely blow up. Whether it looks smooth or breaks is nested scale of that material and its overriding properties, at that length. Plug in any real fluid and the tool gives you the exact factual smoothness and breaking points you were seeking.

The equation

It is not a PDE. It is the molecular floor plus Lemma G, then the tool.

Floor. The substance has a smallest hydrodynamic length $\sigma_* > 0$ — the shortest length at which talking about a flow velocity still means anything. Below σ_* there is no wind. Lemma F and Theorem 1 compute it from the material.

No blow-up. Finitely many particles, a core $\Phi(r) \rightarrow +\infty$ as two particles meet, finite energy: no two occupy one point, no particle becomes infinite. That is Lemma G, stated and proved in **Smooth or break at that length**.

Tool. Name the molecule. Sit at a length L and a speed U . The working law is the first gate that fires, written out in **How to run it**. That winner is smooth or break for that material at that length.

Pins are the numbers of that substance: viscosity, density, molecular size or mean free path, cohesion. A **tile** is a bundle of particles whose mass, momentum, and kinetic energy add. A **gate** is which force wins at the length you are sitting at. The **nest** is that stack of lengths for one material.

Plug the pins of any real fluid into that. That is the equation the bold sentence promised. The rest of the paper proves it.

1. The Fefferman statement

The 2000 statement of Fefferman is a yes-or-no with two boxes.

Box A — always smooth: for every nice start, u and p stay smooth on $\mathbb{R}^3 \times [0, \infty)$.

Box B — blow-up: there exists a nice start for which that field becomes unbounded in finite time.

The unknown is a velocity $u(x, t)$ and a pressure $p(x, t)$. Incompressible. One viscosity $\nu > 0$. The domain is all of $\mathbb{R}^3$. Clay names water and air as examples of this field, not as two different materials. Fefferman does not ask which fluid. He asks you to tick A or B for that field — one tick, all fluids.

What the fight is, inside the sentence: inertia $(u \cdot \nabla)u$, quadratic, versus viscosity $\nu \nabla^2 u$, linear, at every length, with no floor. A means viscosity never loses. B means inertia wins at a point of the field.

Viscosity names a substance

ν is kinematic viscosity: a measured number of an actual liquid or gas. You cannot write a ν without that number belonging to some substance. That substance has other properties the two-term field does not write down — a molecular size or a mean free path, cohesion, density. Those properties do real work.

Air at room pressure: the mean free path is about 6.5×10^{-8} m. Below that length there is no wind. There is a molecule hitting a molecule. Fefferman's sentence still asks for a velocity there. Air does not have one.

Water is not glycerol. Same 1 m, same 1 m/s: water has Reynolds number 10^6 , cascade still running down to $32 \mu\text{m}$. Glycerol has Reynolds number 900, cascade already dead at 6 mm. One tick for both is wrong about at least one of them.

So there is not one A and not one B. Smooth or break is the nest of that substance, at that length. If the field blows up, it has left the fluid.

2. Fluids have a smallest length

A molecular fluid has a smallest scale $\sigma_* > 0$. Below σ_* there is no hydrodynamic velocity. For air that length is a mean free path. For liquid water it is molecular size. You compute it from the material. You do not choose it.

Lemma F (the floor is determined). *If F is a liquid with finite density $\rho > 0$ and finite molar mass $M > 0$, then $n = \rho N_A / M > 0$ and $\sigma_* = n^{-1/3} > 0$. If F is a gas with finite number density $n > 0$ and collision diameter $d > 0$, then $\sigma_* = 1 / (\sqrt{2} \pi d^2 n) > 0$. In both cases the floor is computed from pins of F . It is not chosen.*

Proof. N_A is a positive constant. The two formulae are the liquid spacing and the kinetic-theory mean free path. Both are positive whenever the pins are. ■

Theorem 1. *A molecular fluid has $\sigma_* > 0$. Below σ_* there is no hydrodynamic velocity.*

Proof of Theorem 1. Lemma F. Below σ_* the length is a molecule or a free path. There is no continuum parcel, so there is no hydrodynamic velocity. ■

Fefferman's sentence asks for a velocity at every length, including lengths smaller than σ_* . That quantity is not a property of the fluid.

Chapman–Enskog is the kinetic-theory derivation that a continuum velocity is hydro only when the Knudsen number is small: mean free path much smaller than the length you asked ($\text{Kn} \ll 1$). Beale–Kato–Majda is the vorticity test for breakdown, written for 3-D Euler and read the same way on this field: the time-integral of maximum vorticity diverges, or it does not.

Theorem 2. *Incompressible Navier–Stokes asks for a velocity at every length. Therefore it asks for quantities the fluid does not have. At scales $\gg \sigma_*$ the field is an effective theory (Chapman–Enskog, $\text{Kn} \ll 1$). If that effective theory develops a singularity, it has left its regime.*

Proof of Theorem 2. The unknown of incompressible Navier–Stokes is a velocity on all of $\mathbb{R}^3$, every length. Theorem 1: lengths below σ_* are not a velocity of F . Chapman–Enskog: the field is hydro only for $\text{Kn} \ll 1$. If the field becomes unbounded, or Beale–Kato–Majda diverges, the cascade has left that regime. That solution is not F at that time. ■

So the field is not the fluid. It was written past the floor, and it was written as if every material were the same. Infinite zoom is a property of the field. Nested strengths are a property of the substance.

Lemma V (viscosity is never bare). *Fefferman’s ν is kinematic viscosity: a measured momentum diffusivity of a substance, not an arbitrary scalar. The two-term field was derived from conservation on a continuum approximation of real matter. The terms carry physical meaning by construction, not by choice. A bare number for ν omits the other properties of that substance — floor, cohesion, and the rest. Those omitted properties do real work. If $u(x, t)$ is implicitly that matter, it inherits them whether written or not. Every physically possible instantiation of ν comes attached to a substance that violates the continuum assumption at some length. The object in his sentence — a field with a well-defined velocity at every length — cannot correspond to anything physically real.*

Proof. Kinematic viscosity is $\nu = \mu/\rho$, measured of a material. Lemma F: that material has $\sigma_* > 0$. Theorem 1: below σ_* there is no hydrodynamic velocity. Chapman–Enskog: continuum hydro is $\text{Kn} \ll 1$. So any real ν already names a substance whose continuum dies somewhere. A field asked to exist at every length is not that substance. ■

3. One tick cannot cover two fluids

Box A and Box B classify one blank field. They do not classify fluids.

Water, air, glycerol, and mercury at the same 1 m, 1 m/s are printed in **Same flow, different fluids**. They do not share a nest.

If you tick A, you have also claimed the field stays smooth at lengths the fluid does not have.

If you tick B, you have claimed a blow-up of the field. That is the field leaving its regime (Theorem 2). It is not a molecule becoming infinite.

ν is a substance (Lemma V). Water is not glycerol: different pins, different equation at that length. A second molecule disproves the tick.

Lemma M (a second molecule disproves the tick). *Let a solution of Fefferman's formula be a single tick, A or B, claimed for the two-term field and therefore for all fluids. Let F and F' be two molecular fluids whose pins differ — viscosity, floor, or cohesion. Then there is a flow (U, L) at which the nest of F and the nest of F' disagree. The tick is false of at least one of them. Any such solution is disproved by attaching a second molecule.*

Proof. Lemma V: ν is a substance, not a free parameter. Viscosity, σ_* , and cohesion are pins of the molecule (Lemma F, and $\nu = \mu/\rho$ of the substance). They enter $\text{Re} = UL/\nu$, $\eta = L\text{Re}^{-3/4}$, $\text{Kn} = \sigma_*/L$, and the cohesion ratio. Those are the equation at that level. Change the molecule, change the pins, change the equation, and any proof that used the old pins is not a proof about the new fluid. The tables in **Same flow, different fluids** print the case: at the same $U = 1$ m/s and the same $L = 1$ m, water has $\text{Re} = 10^6$ with its cascade reaching $32 \mu\text{m}$, while glycerol has $\text{Re} \approx 900$ with its cascade already dead at 6 mm. Tick A for both is false of at least one. Tick B for both is false of at least one. ⁴He below 2.17 K has superfluid viscosity 0: it is not even in the $\nu > 0$ box. ■

4. Smooth or break at that length

The right question is not “A or B for all fluids?” It is: at this length, for this material, which strength wins?

That is determined. Every material has a stack of forces. At each length, one of them wins. Whether it looks smooth or breaks is that winner, not a universal tick.

A fluid is particles, then tiles of particles, then tiles of tiles. Mass, momentum, and kinetic energy of the whole are the nested sums (Theorem 3). The live force is the gate that wins at that tile. At hydro tiles, inertia versus viscosity can run, and which one wins is Re , which is material. At σ_* the core is the override. In ordinary flow the cascade already dies well above the floor, and how far above is a property of the material (water $\simeq 10^5$ times the floor, air $\simeq 4 \times 10^3$, glycerol $\simeq 10^7$, mercury $\simeq 2 \times 10^4$). The fluid cannot run to infinite zoom. It also cannot share one A/B with a different fluid.

The hydro gate is $\text{Re} = UL/\nu$. Kolmogorov length $\eta \simeq L\text{Re}^{-3/4}$. When the tile side reaches σ_* , the working law is no longer $(u \cdot \nabla)u$ vs $\nu \nabla^2 u$.

Water at 20 °C, $\nu \approx 10^{-6} \text{ m}^2/\text{s}$, $\sigma_* = 3.104 \times 10^{-10} \text{ m}$:

flow	Re	η	η/σ_*
1 m at 1 m/s	10^6	3.16×10^{-5} m	1.02×10^5
1 m at 10 m/s	10^7	5.62×10^{-6} m	1.81×10^4

Air at STP, $\nu \approx 1.5 \times 10^{-5}$ m²/s, floor = mean free path $\approx 6.5 \times 10^{-8}$ m, 1 m at 1 m/s: $\text{Re} \approx 6.67 \times 10^4$, $\eta \approx 2.41 \times 10^{-4}$ m, $\eta/\sigma_* \approx 3.71 \times 10^3$. Ordinary turbulence dumps the cascade above the floor. Blow-up of the field would be that cascade refusing to stop. The nest stops it.

The floor is steep. Lennard-Jones core $\simeq (\sigma_*/r)^{12}$ against attraction $\simeq (\sigma_*/r)^6$. At $r = 2\sigma_*$ the 12-term is 2.44×10^{-4} . At $r = 0.95\sigma_*$ it is 1.85. At $r = \sigma_*/2$ it is 4096. Flat, then a wall.

Lemma G (the object has no blow-up). *Let $N < \infty$ particles interact by a pair potential Φ with $\Phi(r) \rightarrow +\infty$ as $r \rightarrow 0$ (a core), and let the total energy be finite. Then $\inf_{i \neq j} r_{ij}(t) > 0$ for the life of the system. No two particles occupy one point. No particle becomes infinite. A blow-up of $\|u\|_\infty$ for a continuum field is not this.*

Proof. Energy $E = K + \sum_{i < j} \Phi(r_{ij})$ is conserved, or at least bounded if a thermostat is present. Kinetic energy $K \geq 0$, so the sum of Φ is bounded above by that bound. If some pair had $r \rightarrow 0$, then $\Phi \rightarrow +\infty$, which contradicts the bound. The same bound caps K , so every speed is bounded (finite masses). Bounded speed, finite time: no particle reaches spatial infinity. ■

A gas floor is the same fact at a larger σ_* : collision diameter, then mean free path. Superfluid helium is not in Fefferman's $\nu > 0$ box at all, and the tool still returns a verdict for it: at that level, that substance is not described by the two-term field.

Theorem 2'. *There is not one smooth-or-blow-up for all fluids. The winner is the nest of that material, at that level. No molecular fluid blows up at infinite zoom: if the two-term field blows up, it is no longer describing the fluid.*

Proof of Theorem 2'. Lemma M: one tick for the blank field is already false once two molecules exist. Theorem 1: every F has a floor, so F has no infinite zoom. Re and σ_* are pins of F , so at a fixed (U, L) the winner depends on the material. Lemma G: F does not blow up. Theorem 2: if the two-term field blows up, it is not F . Therefore there is not one smooth-or-blow-up for all fluids, and no molecular fluid blows up at infinite zoom. ■

Fefferman's unknown was a velocity at every x . That is not a property of the fluid (Theorem 2). The unknowns of the object are tile totals of the actual particles. Theorems 3 and 4 are that representation.

Theorem 3. *Let a level of the nest be a partition of the particles into disjoint tiles τ_j . Write*

$$\begin{aligned} M(\tau) &= \sum_{i \in \tau} m_i \\ P(\tau) &= \sum_{i \in \tau} m_i v_i \\ K(\tau) &= \sum_{i \in \tau} \frac{1}{2} m_i |v_i|^2 \end{aligned}$$

Then $M(\text{whole}) = \sum_j M(\tau_j)$, $P(\text{whole}) = \sum_j P(\tau_j)$, $K(\text{whole}) = \sum_j K(\tau_j)$. Identities of a partition, not approximations. Front: particles $\rightarrow$ tile totals $\rightarrow$ whole. Back: the whole's (M, P, K) recovered from the tile list by those sums, uniquely. Direct check, 1000 equal-mass particles, ten tiles of 100: relative error 2.3×10^{-15} on P , 1.2×10^{-15} on K — a machine check of the identity, not a physical measurement. The inverse does not recover the particles. That is the point of a tile.

Proof of Theorem 3. The tiles partition the particles. $M(\text{whole}) = \sum_i m_i = \sum_j \sum_{i \in \tau_j} m_i = \sum_j M(\tau_j)$. The same regrouping for P and K . Unique recovery of the totals from the tile list is that identity, read backwards. ■

Temperature of a tile is its peculiar kinetic energy — what is left after the tile's bulk drift is subtracted, $K(\tau) - |P(\tau)|^2 / (2M(\tau))$ — over $3k_B/2$ per molecule. Inverting that gives the tile's peculiar K back, not each v_i .

Pair potential is not strictly additive. Pairs inside a tile stay in the tile. Pairs that straddle a face are the halo. In 3D each tile has at most 26 neighbours. Working law at a level: intra-tile pairs plus moment exchange with the halo, where a moment exchange is one tile passing its totals to a neighbour rather than every pair being computed across the face.

Theorem 4. *For N particles, naive pair count is $N(N-1)/2$. Nested at branching k :*

$$W(k) = \frac{N(k-1)}{2} + 26 \frac{N}{k}$$

Proof of Theorem 4. Tiles of k molecules, N/k tiles. Intra-tile pairs: $(N/k) \cdot k(k-1)/2 = N(k-1)/2$. Halo: at most 26 neighbours per tile, $26N/k$. Sum is $W(k)$. ■

Theorems 3 and 4 are the representation: tile totals instead of a value at every point, and bounded work instead of every pair, with the floor of Theorem 1 underneath.

5. Same flow, different fluids

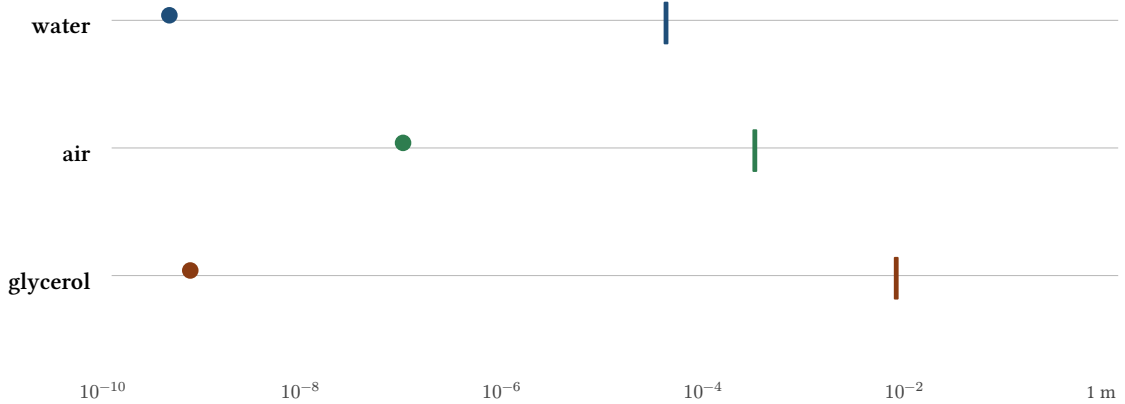
Real fluids do not share a nest. Viscosity, floor, cohesion, and where the continuum dies are numbers of the substance.

The adding-up of Theorem 3 works for every fluid. What moves when those numbers move is which gate wins, and at which tile.

Theorem 3'. *Let Q be any per-particle quantity that adds, so $Q(\tau) = \sum_{i \in \tau} Q_i$. Then M , P , K and Q are all tile totals of a partition, by the same regrouping as Theorem 3. Quantities derived from those totals follow: number density $n = M/V$, temperature from peculiar K , boundness from cohesion $/k_B T$. Water, air, glycerol, mercury and plasma obey that same additive rule.*

Proof of Theorem 3'. Theorem 3 is the case $Q \in \{M, P, K\}$. For any other per-particle Q the regrouping is the same: $\sum_i Q_i = \sum_j \sum_{i \in \tau_j} Q_i$, because the tiles partition the particles. The derived quantities are functions of those totals and the tile volume, so they are fixed once the totals are. Which gate wins is not part of the sum; it is read off the ratios at that length, in the tables below. ■

Pins at 20 °C except air at STP. Hydro comparison is 1 m at 1 m/s. Air's floor is the mean free path (the gas has already unbound). Liquids' floor is molecular size.



Dot = floor σ_* . Bar = Kolmogorov η at 1 m, 1 m/s. Right edge = the metre. Same flow, three materials, three nests.

Figure 1: Same 1 m at 1 m/s. Water dumps the cascade far above a molecule. Air's floor is already a mean free path. Glycerol's cascade dies at millimetres. One blank field cannot be all three.

fluid	ν (m ² /s)	floor σ_*	Re	η	η/σ_*	what it is
water	1.00×10^{-6}	3.10×10^{-10} m	1.00×10^6	3.16×10^{-5} m	1.02×10^5	H-bond liquid

fluid	ν (m ² /s)	floor σ_*	Re	η	η/σ_*	what it is
air	1.50×10^{-5}	6.5×10^{-8} m (mfp)	6.67×10^4	2.41×10^{-4} m	3.71×10^3	gas; continuum dies at mfp
glycerol	1.12×10^{-3}	$\simeq 5 \times 10^{-10}$ m	8.94×10^2	6.12×10^{-3} m	1.22×10^7	high- ν liquid; cascade dies millimetres up
mercury	1.12×10^{-7}	$\simeq 3.0 \times 10^{-10}$ m	8.90×10^6	6.14×10^{-6} m	2.05×10^4	liquid metal; low ν , metallic cohesion

Continuum break is Knudsen $\text{Kn} = \sigma_*/L \simeq 0.1$, a nest level, not a PDE event.

fluid	L where $\text{Kn} = 0.1$	what breaks
water	$\simeq 3 \times 10^{-9}$ m	still molecular; hydro tiles are fine at microns
air	$\simeq 6.5 \times 10^{-7}$ m	continuum already dead at sub-micron, $200 \times$ coarser than water
glycerol	$\simeq 5 \times 10^{-9}$ m	like water, but viscosity won much earlier ($\eta = 6$ mm)
mercury	$\simeq 3 \times 10^{-9}$ m	like water in size, unlike water in Re

Thermal unbind is boiling, a material pin, not a universal field:

fluid	bound at 300 K	break pin
water	liquid; $\Delta H_{\text{vap}}/k_B T \approx 16.3$	boils 373 K
air (N ₂)	already gas; $\Delta H_{\text{vap}}/k_B T \approx 2.23$	boiled at 77 K
glycerol	liquid; $\Delta H_{\text{vap}}/k_B T \approx 36.5$	boils 563 K
mercury	liquid metal; $\Delta H_{\text{vap}}/k_B T \approx 23.7$	boils 630 K
⁴ He	Fefferman's $\nu > 0$ field does not apply	superfluid below 2.17 K, superfluid viscosity 0
plasma	EM and centrifugal gates, not H-bond	E_{cloud} , The same tool on plasma

Superfluid helium is not in Fefferman's $\nu > 0$ field at all. Plasma is not incompressible Navier–Stokes; it is the cloud in **The same tool on plasma**.

Showing several fluids is the point. The nest is the object. The blank field cannot see the bind or the break.

The first rows below are ordinary fluids. Plasma rows are inverted later. Nuclear forces are outside this paper.

Gate	ratio	who wins, where
Electromagnetic / gravity (two protons)	1.24×10^{36}	EM, molecule to mountain. Gravity is not the lab-fluid law.

Gate	ratio	who wins, where
Electromagnetic / gravity (two electrons)	4.17×10^{42}	same
One H-bond / thermal (20 kJ/mol vs $k_B T$)	8.02 at 300 K, 6.45 at 373 K	cling still wins for liquid water; heat is climbing
Self-gravity of 1 m ³ of water / Earth's g	1.77×10^{-8}	inertia and Earth-weight, not self-gravity
Inertia / viscosity	$\text{Re} = UL/\nu$	hydro tiles; glycerol vs mercury at the same U, L
Viscous cascade / floor	η/σ_*	water $\simeq 10^5$, air $\simeq 4 \times 10^3$, glycerol $\simeq 10^7$, mercury $\simeq 2 \times 10^4$
Core / attraction	$(\sigma_*/r)^{12}$ vs $(\sigma_*/r)^6$	override at the molecule
Shear / MHD growth	$\omega_{E \times B}/\gamma_{\max}$	A3
Pressure / magnetic+centrifugal	A4	outer shell: centrifugal already the override
Centrifugal / thermal	A1 exponent	inverted for ω
Nuclear	femtometre nest	outside this paper

Density, temperature, speed, impact, and electromagnetic field strength are the inputs to those ratios. Change a variable far enough and the winner swaps.

6. How to run it

You have a material and a question at a length L , usually with a speed U . Sit that material at that length. The working law is the gate that wins there. The output is a verdict — which law, whether a continuum field is even allowed, and the pins recovered by inverting that law — not a function $u(x, t)$ on all of space. You still get closed formulas once the nest has picked the law. They are formulas at one level, not at every level: the Kolmogorov length η is where the turbulent cascade dumps its energy, and the plasma law A2 in **The same tool on plasma** is an ordinary differential equation on one chosen gate. Neither asks for a velocity at every length.

Floor, tile, gate, invert:

tool	object	what you compute
Floor	$\sigma_* > 0$	stop length: below it there is no hydrodynamic velocity
Tile	$M(\tau), P(\tau), K(\tau)$, halo	additive totals of a partition; work $W(k)$, not $N(N-1)/2$
Gate	ratio of two strengths, at this L	the winner is the working law at that level
Invert	front and back of that law	recover the pin from the output, uniquely

How to pin a material you have not seen in this paper. Look up kinematic viscosity ν (or μ and ρ , then $\nu = \mu/\rho$), density ρ , molar mass M , temperature T , and molar enthalpy of vaporization ΔH_{vap} . CODATA: $k_B = 1.380649 \times 10^{-23}$ J/K, $N_A = 6.02214076 \times 10^{23}$ mol $^{-1}$.

Liquid floor: $n = \rho N_A / M$, then $\sigma_* = n^{-1/3}$. Water at 1000 kg/m 3 , $M = 18.01528 \times 10^{-3}$ kg/mol, gives $\sigma_* = 3.104 \times 10^{-10}$ m, as in **Counting the nest**.

Gas floor: number density $n = P/k_B T$, collision diameter d , then $\sigma_* = 1/(\sqrt{2} \pi d^2 n)$, the mean free path. Air at STP with $d \approx 3.7 \times 10^{-10}$ m gives $\approx 6.5 \times 10^{-8}$ m, the pin in **Same flow, different fluids**. If you already have a measured mfp, that is σ_* .

Cohesion: $(\Delta H_{\text{vap}}/N_A)/k_B T$. That is the whole-molecule ratio, the energy it takes to remove one molecule from the liquid against $k_B T$. Water at 300 K is ≈ 16.3 . The single-bond row in **Same flow, different fluids** is the same comparison for one H-bond alone, ≈ 8 ; a water molecule holds more than one bond, which is why the whole-molecule number is larger. If cohesion has fallen through ≈ 1 , or T has passed the boil pin, the liquid has unbound. ^4He below 2.17 K has superfluid viscosity 0: Fefferman's $\nu > 0$ field is not the object.

Tile. The unknowns at a level are tile totals, not a field value at every x :

$$M(\tau) = \sum_{i \in \tau} m_i$$

$$P(\tau) = \sum_{i \in \tau} m_i v_i$$

$$K(\tau) = \sum_{i \in \tau} \frac{1}{2} m_i |v_i|^2$$

The whole is the sum of the tiles (Theorem 3). Temperature of a tile is its peculiar kinetic energy — what is left after the tile's bulk drift is subtracted, $K(\tau) - |P(\tau)|^2 / (2M(\tau))$ — over $3k_B/2$ per molecule. Pairs inside stay in the tile. Pairs that straddle a face are the halo: in 3D at most 26 neighbours.

To nest: pick branching $k \in \{10, 20, 100\}$. Level 0 is one molecule, side σ_* . Each level multiplies molecule count by k and side by $k^{1/3}$. Stop going up when the tile side reaches the L in the question. Stop going down at σ_* . Work at a level is intra-tile pairs plus moment exchange with the halo:

$$W(k) = N(k-1)/2 + 26N/k$$

not $N(N-1)/2$.

Gates live at lengths. Do not dump every ratio in the paper into one list and pick the biggest number. Re at 1 m does not compete with a Lennard-Jones 12-term at 0.3 nm. Sit at L , then compute only the rows that act there.

At this L , the formulas are:

$$\text{Re} = UL/\nu$$

$$\eta = L \text{Re}^{-3/4}$$

$$\text{Kn} = \sigma_*/L$$

η is the length where a turbulent cascade would dump its energy, and it only means that if Re is large enough to give $\eta < L$. Also compute η/σ_* , cohesion $(\Delta H_{\text{vap}}/N_A)/k_B T$, and, if r is near σ_* , core / attraction $(\sigma_*/r)^{12}$ vs $(\sigma_*/r)^6$. If the machine has B, ω, n , those rows too (A3, A4 in **The same tool on plasma**).

Walk the stops in this order. First one that fires is the working law.

1. The material is not a single Newtonian fluid with one $\nu > 0 \rightarrow$ Fefferman's field is not the object. Write the actual substance (^4He , plasma, ...).
2. $\text{Kn} \simeq 0.1$ or larger $\rightarrow$ continuum is dead (the pin in **Same flow, different fluids**). Working law: tiles at the floor, not Navier–Stokes.
3. Cohesion unbound (boil pin, or cohesion ≈ 1 , or already a gas) $\rightarrow$ not a clinging liquid.
4. $L \approx \sigma_*$ $\rightarrow$ hydro is gone. Core vs attraction. The 12-term is the override (flat, then a wall).
5. $\text{Kn} \ll 1$ and $L \gg \sigma_*$ $\rightarrow$ hydro tile. If $\eta \geq L$, there is no inertial cascade: viscosity already won at L . If $\eta \ll L$ and $\eta \gg \sigma_*$, ordinary turbulence: the cascade dumps above the floor. If a formula gave $\eta \leq \sigma_*$, the two-term field has left the fluid; the floor has the override. At this same hydro tile, Re says inertia vs viscosity: glycerol at 1 m, 1 m/s is $\text{Re} \approx 900$ (viscosity); mercury at the same flow is $\text{Re} \approx 9 \times 10^6$ (inertia).
6. An extra gate (centrifugal, magnetic, ...) larger than the hydro row $\rightarrow$ that gate has the tile.

Whichever stop fires, that is a change of level inside the nest. It is not a molecule becoming infinite, and it is not Box B.

Write the verdict as one line, where “NS allowed?” asks whether Navier–Stokes is a legitimate description at that length:

material, $T \mid U, L \mid \sigma_* \mid \text{Re} \mid \eta \mid \eta/\sigma_* \mid \text{Kn} \mid \text{winner} \mid \text{NS allowed?} \mid \text{working law}$

Then invert that law and check you get the pin back. If you cannot recover it, one of the material's numbers is wrong or missing.

law	front	back
Re	$\text{Re} = UL/\nu$	$U = \text{Re } \nu/L, L = \text{Re } \nu/U$
η	$\eta = L \text{Re}^{-3/4}$	$\text{Re} = (L/\eta)^{4/3}$
Kn	$\text{Kn} = \sigma_*/L$	$L = \sigma_*/\text{Kn}$
A1	$R(\omega) = \exp(m\omega^2 \Delta_r / (2k_B T))$	$\omega(R) = \sqrt{\ln(R) \cdot \frac{2k_B T}{m\Delta_r}}, R > 1$

A2, A3, A4 invert in **The same tool on plasma**. Direct check on A1: $R = 15$ returns ω , that ω returns 15 at relative error 1.3×10^{-15} .

If you need another length, change L , retile, repeat from the stops. Do not copy $(u \cdot \nabla)u$ versus $\nu \nabla^2 u$ through the floor. For one fluid at one length, the answer is the winner at that level.

Worked problem – water in a 1 cm pipe at 1 m/s. Pins from this paper: $\nu = 1.00 \times 10^{-6} \text{ m}^2/\text{s}$, $\sigma_* = 3.104 \times 10^{-10} \text{ m}$, $T = 293 \text{ K}$, cohesion ≈ 16 . Still a liquid. $L = 0.01 \text{ m}$, $U = 1 \text{ m/s}$.

Re	η	η/σ_*	Kn
1.00×10^4	$1.00 \times 10^{-5} \text{ m}$	3.22×10^4	3.10×10^{-8}

$\text{Kn} \ll 0.1, L \gg \sigma_*, \eta < L, \eta/\sigma_* \gg 1$. Hydro is allowed as an effective theory at 1 cm. Cascade dumps four orders above the floor. Back: $U = \text{Re } \nu/L = 1 \text{ m/s}$. Verdict: water, 20 °C | 1 m/s, 1 cm | winner = hydro (inertia/viscosity) | NS allowed at this L | floor still forbids infinite zoom.

Worked problem – air at a micron. Pins: $\nu = 1.50 \times 10^{-5} \text{ m}^2/\text{s}$, $\sigma_* = 6.5 \times 10^{-8} \text{ m}$ (mfp). $L = 1.0 \times 10^{-6} \text{ m}$, $U = 1 \text{ m/s}$.

Re	η	η/σ_*	Kn
6.67×10^{-2}	$7.62 \times 10^{-6} \text{ m}$ ($\eta > L$: no cascade)	1.17×10^2	0.065

Kn is already ≈ 0.1 . Continuum is dying. Stop at step 2. Do not evolve Navier–Stokes at this L . Working law: tiles from the mean free path. Air's continuum is already dead at $L \approx 6.5 \times 10^{-7} \text{ m}$ ($\text{Kn} = 0.1$ in **Same flow, different fluids**). Verdict: air, STP | 1 m/s, 1 μm | winner = Knudsen / floor | NS not allowed.

A machine with B, ω, n runs the same way: inner and outer zones are two tiles; A1–A7 are the gates; invert as in **The same tool on plasma**. The tables in **Same flow, different fluids** are this recipe on water, air, glycerol, mercury, and ^4He at 1 m, 1 m/s. **Counting the nest** is the nest count for 1 m^3 of water.

The tool is four steps: floor, tile, gate, invert.

Every molecular fluid

The job was: does a fluid stay smooth or blow up? Fefferman wrote that as one tick for a two-term field on all of space, with Beale–Kato–Majda as the vorticity test for blow-up on it.

The object is a molecular fluid F : finitely many particles, a core $\Phi(r) \rightarrow +\infty$ as $r \rightarrow 0$ (Lennard-Jones is this Φ), or a gas whose floor is a mean free path. Water, air, glycerol, mercury, helium, and plasma are cases of F .

ν is never bare (Lemma V). A tick of the blank field is false as soon as two molecules disagree (Lemma M). Theorems 1 and 2 are why that field cannot return a fact about F . The tool returns that same information — smooth or break — but about the molecular fluid instead of the blank field.

Lemma S (the tool is a function). *The stops in **How to run it** are a listed order. The first stop that fires is unique. For the question whether F looks smooth or breaks at (L, U) , under the forces named in this paper, every case hits exactly one working law: not Fefferman’s $\nu > 0$ field at all; or continuum dead ($\text{Kn} \simeq 0.1$); or cohesion unbound; or $L \approx \sigma_*$ (core); or a hydro tile ($\text{Kn} \ll 1, L \gg \sigma_*$), partitioned by $\eta \geq L, \eta \ll L$ and $\eta \gg \sigma_*$, or $\eta \leq \sigma_*$; or an extra named gate whose ratio outranks the hydro row. $\text{Kn} \simeq 0.1$ is the conventional kinetic-theory edge of the $\text{Kn} \ll 1$ condition in Theorem 2, not a free parameter.*

Proof. An ordered list has a unique first true item. The hydro three-way on η is a partition of that tile. The femtometre nest of a nucleus is outside the forces named here, so it is not an undecided row of this question. If inverting the winning law does not return the pin you started from, one of the material’s numbers is wrong or missing; each law in the invert table has exactly one inverse. ■

Theorem 5. *For every molecular fluid F , and every question asked at a length L with a speed U :*

$\sigma_(F) > 0$ and is determined (Theorem 1, Lemma F). F itself has no blow-up (Lemma G). A hydrodynamic velocity is only a property of F at scales $\gg \sigma_*$ (Theorem 2).*

The working law at (F, L, U) is the first stop that fires in the tool of this paper; that law is the unique winner of the nest at that level (Theorem 2’, Lemma S). Whether F looks smooth or breaks at that level is that winner. It is determined.

Infinite zoom is not a property of F . If a solution of the two-term field becomes unbounded, or if Beale–Kato–Majda diverges on that field, that solution is not F at that time (Theorem 2, Lemma G).

Proof. Lemma V: ν is never bare. Lemma M: a tick for all fluids is false as soon as two molecules disagree, and they do. Lemma F and Theorem 1: every F has a determined floor. Lemma G: every core with finite energy has no particle blow-up. Theorem 2: past σ_* the two-term field is not F ; a singularity of the field is leaving the regime. Theorem 2': the winner is the nest of that material, at that level. Theorems 3, 3', 4: tiles and nested work are the representation. Lemma S: **How to run it** is those theorems used as a tool, and the verdict is unique. Smooth or break was the information the 2000 statement asked for. Name the molecule, run the tool. That is the verdict. ■

You do not tick one box for all fluids. You run the tool on the fluid you have. That is how you get what the 2000 statement asked for.

7. Counting the nest

Here is 1 m^3 of water, counted. $m = 2.991508 \times 10^{-26} \text{ kg}$, $n = 3.342796 \times 10^{28} \text{ m}^{-3}$. One molecule: $(1/n)^{1/3} = 3.104 \times 10^{-10} \text{ m}$. Branching $k \in \{10, 20, 100\}$. Level 0 is one molecule. Level ℓ is k^ℓ molecules, until the tile side reaches the length in the question. The top row is capped there: 100^{15} would overshoot the metre, so level 15 is the N molecules of the whole cube.

$k = 100$, 1 m^3 of water, 15 levels:

level	molecules in a tile	tile side	tiles in 1 m^3
0	1	$3.104 \times 10^{-10} \text{ m}$	3.343×10^{28}
1	10^2	$1.441 \times 10^{-9} \text{ m}$	3.343×10^{26}
2	10^4	$6.688 \times 10^{-9} \text{ m}$	3.343×10^{24}
5	10^{10}	$6.688 \times 10^{-7} \text{ m}$	3.343×10^{18}
10	10^{20}	$1.441 \times 10^{-3} \text{ m}$	3.343×10^8
15	3.343×10^{28}	1 m	1

$k = 10$ is 29 levels to the same metre. $k = 20$ is 22.

$N = 3.343 \times 10^{28}$:

naive	$W(10)$	$W(20)$	$W(100)$
5.587×10^{56}	2.373×10^{29}	3.610×10^{29}	1.663×10^{30}

Naive $/W(100) = 3.36 \times 10^{26}$. Using moment exchange only, at tile side L : work = $27(1 \text{ m} / L)^3$.
 At 1 mm: 2.7×10^{10} . At 1 μm : 2.7×10^{19} . At 1 m: 27.

Plasma tiles. Inner $n = 5.000 \times 10^{19} \text{ m}^{-3}$, outer at $R = 15$: $n = 7.500 \times 10^{20} \text{ m}^{-3}$. Mean ion spacing inner: $2.714 \times 10^{-7} \text{ m}$. A 100-ion tile: $1.260 \times 10^{-6} \text{ m}$. Nest below the two zones: ions $\rightarrow 10 \rightarrow 100 \rightarrow \dots$ until the zone.

8. The same tool on plasma — E_{cloud}

A gate is which force wins at this length. Water and air showed that above: inertia versus viscosity, then the molecular floor. This section is the same idea on a rotating plasma, so the nest is not something that only works for water. A1 through A7 are seven named balances. Invert means solve the same formula backwards. You can skip the plasma and still have the proof from water, air, glycerol, and mercury. The plasma is the check that the tools still invert on another material.

The material is a two-zone rotating plasma, from the author's reactor manuscript of 17 February 2026. Annulus $r_{\text{inner}} = 0.2 \text{ m} < r < r_{\text{outer}} = 1.0 \text{ m}$.

Pins: $m = 3.34 \times 10^{-27} \text{ kg}$, $T = 1.7 \times 10^8 \text{ K}$ (15 keV), $n_{\text{core}} = 5 \times 10^{19} \text{ m}^{-3}$, $\nu_{\text{col}} = 10^6 \text{ s}^{-1}$, $\gamma_{\text{max}} = 10^5 \text{ s}^{-1}$, $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$, $k_B = 1.380649 \times 10^{-23} \text{ J/K}$.

A1.

$$n(r) = n_{\text{core}} \exp\left(\frac{m\omega^2(r^2 - r_{\text{inner}}^2)}{2k_B T}\right)$$

(solid-body ω , uniform T , $r_0 = r_{\text{inner}}$)

A2.

$$r'' + \nu_{\text{col}} r' - \omega^2 r = 0$$

A3.

$$\omega_{E \times B} = \left| (r/q) d/(dr) (qv_\phi/r) \right| > \gamma_{\text{max}}$$

, $v_\phi = \omega r$

A4.

$$nk_B T \leq \frac{B^2}{2\mu_0} + \frac{1}{2} nm\omega^2 r^2$$

A5.

$$\Gamma_{\text{in}}(r_{\text{inner}}) = 5 \times 10^{21} \text{ s}^{-1}$$

only

A6.

$$d(\nabla n)/(dt) = -0.01 dT/(dt)$$

$$\textbf{A7. } n_{\text{outer}} T_{\text{outer}} \tau_E \geq 10^{21} \text{ keV}\cdot\text{s}/\text{m}^3$$

A1 bijection. Fix m, T , radii. Write $\Delta_r = r_{\text{outer}}^2 - r_{\text{inner}}^2 = 0.96 \text{ m}^2$ for the radial span.

Front:

$$R(\omega) = \exp\left(\frac{m\omega^2 \Delta_r}{2k_B T}\right)$$

Back:

$$\omega(R) = \sqrt{\ln(R) \cdot \frac{2k_B T}{m\Delta_r}} \quad (R > 1)$$

R_{target}	$\omega \text{ (s}^{-1}\text{)}$	$R(\omega)$	relative error
10	1.836031×10^6	10.0000000000	1.1×10^{-15}
15	1.991135×10^6	15.0000000000	1.3×10^{-15}
20	2.094227×10^6	20.0000000000	3.6×10^{-16}

The reactor manuscript also lists $v = 2 \times 10^5 \text{ m/s}$ at r_{inner} , which is a different operating point: $\omega = 1 \times 10^6 \text{ s}^{-1}$, $R = 1.9799166055$. Invert recovers that ω at relative error 0. The 10–20 $\times$ compression shell is ω in $[1.836 \times 10^6, 2.094 \times 10^6] \text{ s}^{-1}$, not that 1×10^6 value.

A2 closed form. $s_{\pm} = \left(-\nu_{\text{col}} \pm \sqrt{\nu_{\text{col}}^2 + 4\omega^2}\right)/2$. $r(t) = Ae^{s_+ t} + Be^{s_- t}$, $r(0) = r_{\text{inner}}$, $r'(0) = 0$. At $R = 15$: $\tau = 1.340764 \times 10^{-6} \text{ s}$, $r(0) = 0.200000000000 \text{ m}$, $r(\tau) = 1.000000000000 \text{ m}$. Invert $\tau \rightarrow \omega$ at relative error 2.3×10^{-16} . The 15–20 μs migration time quoted in that manuscript is not this ODE at these pins; the $\omega = 10^6 \text{ s}^{-1}$ operating point gives 3.13 μs .

A3. Uniform q , uniform ω : $\omega_{E \times B} = 0$, gate fails. Linear $q(r) = 1 + 2.5(r - 0.2)$. At $R = 15$, $\omega_{E \times B}$ from 9.956×10^5 to $1.659 \times 10^6 \text{ s}^{-1}$ on $r \in [0.2, 1.0]$, all above γ_{max} .

r (m)	q	$\omega_{E \times B}$ (s ⁻¹)	$> \gamma_{\max}$
0.2	1.00	9.956×10^5	yes
0.4	1.50	1.327×10^6	yes
0.6	2.00	1.493×10^6	yes
0.8	2.50	1.593×10^6	yes
1.0	3.00	1.659×10^6	yes

A4. $R = 15$, $B = 4$ T, magnetic pressure 6.366×10^6 Pa. Inner: $P = 1.174 \times 10^5$ Pa $\leq$ mag + cent. Outer: $P = 1.760 \times 10^6$ Pa, centrifugal = 4.966×10^6 Pa, already the override. A4 holds with $B = 0$ at the outer radius.

A5. Fuel is fed into the plasma at the inner edge of the annulus, and nowhere else. Ask what happens if it is fed in at the centre instead and these seven laws have no answer, so the tool returns nothing rather than guessing.

A6. In increments, where Δ here means a change in the quantity that follows it. Front: $\Delta(\nabla n) = -0.01\Delta T$. Back: $\Delta T = -\Delta(\nabla n)/0.01$.

A7. $R = 15$, $T = 15$ keV, $\tau_E = 1$ s: $nT\tau_E = 1.125 \times 10^{22}$ keV·s/m³ $\geq 10^{21}$. At 5 s: 5.625×10^{22} .

Force balance. $a_c = \omega^2 r = v^2/r$. At $R = 15$, r_{outer} : $nma_c = nm v^2/r$ at relative error 0. Invert $v = \sqrt{a_c r}$ recovers 1.991135×10^6 m/s at relative error 0.

Inner and outer zones are two tiles. A1 is their density law, already inverted.

9. What this closes

Name the molecule. Run the tool at the length you asked. That is smooth or break for that fluid. Lemmas F, G, V, M, S and Theorem 5 close it. The two boxes cannot cover two fluids (**One tick cannot cover two fluids**). The nest is in **Smooth or break at that length** and **Same flow, different fluids**. The tool is **How to run it**. Water counted in **Counting the nest**. Plasma inverted in **The same tool on plasma**.

10. Not this problem

A logistic map blowing up in float64 is not a fluid blowing up. Inexact iteration of $x \mapsto 4x(1-x)$ in float64 crosses error 0.1 at step 56; $\ln(2^{53})/\ln 2 = 53$. Appendix A. That is chaos of the number format carrying the map, not a singularity of a fluid.

11. Open

The fluids result is Theorem 5. What follows is open questions about the plasma worked in **The same tool on plasma**, not about the fluids result.

A3 on a Grad–Shafranov q , not only the linear interpolant. A2 with a measured ν_{col} if the 15–20 μs figure was empirical rather than this ODE. A wall offset for the annulus is not pinned by the reactor source, so queries that need it stay incomplete. Pair-potential halo for Coulomb at plasma density, not only the 26-neighbour count. The femtometre nest of a nucleus is a separate question and is not treated here.

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Sources

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Chapman, S., and Cowling, T. G. *The Mathematical Theory of Non-Uniform Gases*. Continuum as $\text{Kn} \ll 1$.

Kolmogorov, A. N. The local structure of turbulence in incompressible viscous fluid for very large Reynolds numbers. *Dokl. Akad. Nauk SSSR* 30 (1941) 301–305. $\eta \simeq L\text{Re}^{-\frac{3}{4}}$.

Beale, J. T., Kato, T., and Majda, A. Remarks on the breakdown of smooth solutions for the 3-D Euler equations. *Comm. Math. Phys.* 94 (1984) 61–66. Their blow-up test on the field: $\int \|\omega\|_{\infty}$.

Jones, J. E. On the determination of molecular fields. *Proc. R. Soc. Lond. A* 106 (1924) 463–477. $\text{Core} \simeq r^{-12}$.

CODATA 2018. k_B, N_A, e, G, m_p, m_e .

Material pins at 20 °C except air at STP: water $\nu \approx 10^{-6} \text{ m}^2/\text{s}$, $\rho = 1000 \text{ kg/m}^3$, boil 373 K; air $\nu \approx 1.5 \times 10^{-5} \text{ m}^2/\text{s}$, $\text{mfp} \approx 6.5 \times 10^{-8} \text{ m}$; glycerol $\mu \approx 1.41 \text{ Pa}\cdot\text{s}$, $\rho \approx 1260 \text{ kg/m}^3$, boil 563 K; mercury $\mu \approx 1.52 \times 10^{-3} \text{ Pa}\cdot\text{s}$, $\rho \approx 13534 \text{ kg/m}^3$, boil 630 K; ^4He λ -point 2.17 K.

Author sources. These are the author’s manuscripts used as sources for this paper. Some are unpublished or private. They are still the factual citations.

In this paper the extra axis is scale: which nest is live, which gate is on. The manuscripts below are sources for that machinery, not part of the argument above.

Crowell, R. D. *NP Is a Property of Representation, Not of Problems*. Cognitive Systems Incorporated. Encoding versus object; cost symmetry inside a complete closure; inexact-carrier lifetime (Appendix A here: 53 predicted, 56 measured).

Crowell, R. D. The Self-Stabilizing Fusion Reactor. 17 February 2026. E_{cloud} ; laws A1–A7; two-zone rotating plasma; pins for $m, T, n_{\text{core}}, \nu_{\text{col}}, \gamma_{\text{max}}, \Gamma_{\text{in}}$.

Crowell, R. D. FUSION-MATH. Verbatim extraction of A1–A7 from the reactor paper. Linguistic Physics Engine evidence.

Crowell, R. D. MASTER Geometric Stability Field Theory (GSFT). 5D field; spectrum as dimension (the extra axis: which gate is live).

Crowell, R. D. Universe Math — Base- λ Logarithmic Folding Scale. Nested-scale rule.

Crowell, R. D. universe math insights. Nested-scale reading.

Crowell, R. D. universe math universal metrology key. Anchor quantities used with CODATA.

Crowell, R. D. universe math Earth math translation. Same metrology family.

Crowell, R. D. Tiled overlap multi-body v1. Locked 17 July 2026. Spatial tiles, overlapping halo, bounded in-tile work, neighbour coupling.

Crowell, R. D. Nested world tick v1. Nest of parent/child lanes; properties compose down the nest.

Crowell, R. D. THE-PHYSICS-ENGINE (element engine). Nested law at material scale; the femtometre nest is not treated in this paper.

Crowell, R. D. Universal Containment Theory — Nuclear Many-Body. The femtometre nest; not treated in this paper.

Crowell, R. D. Executive Summary — GSFT/UMOE Framework. 5D / spectrum split.

Crowell, R. D. GSFT Discovery and Replication Guide.

Crowell, R. D. Master Technical Appendix (GSFT).

Crowell, R. D. PHASES OF THOUGHT. Same Lyapunov / nested-scale machinery at another scale.

Crowell, R. D. COGNITIVE PHYSICS.

Crowell, R. D. unified theory of history math. Nested-scale rule $C \geq \Sigma$ as one law at nested scales.

Appendix A — map lifetime

reproduce.py reprints every number in this paper: run `python reproduce.py`. The three appendices below are the parts worth reading on the page. This one is the check behind **Not this problem**: float64 iteration of $x \mapsto 4x(1 - x)$ crosses error 0.1 at step 56, against the predicted $\ln(2^{53})/\ln 2 = 53$.

```
from decimal import Decimal, getcontext
import math
getcontext().prec = 140

def lifetime(n=140):
    x0 = Decimal("0.40000000000000000000000000000001")
    four, one = Decimal(4), Decimal(1)
    xe, xf = x0, float(x0)
    cross = None
    for i in range(n + 1):
        e = float(abs(Decimal(xf) - xe))
        if cross is None and e > 0.1:
            cross = i
        xe = four * xe * (one - xe)
        xf = 4.0 * xf * (1.0 - xf)
    return cross, math.log(1 / 2**-53) / math.log(2)

# measured 56, predicted 53.0
```

Appendix B — A1 bijection and A2 ODE

```
import math
kB, m, T = 1.380649e-23, 3.34e-27, 1.7e8
r_in, r_out, nu = 0.2, 1.0, 1e6
Delta_r = r_out**2 - r_in**2

def omega_from_R(R):
    return math.sqrt(math.log(R) * 2 * kB * T / (m * Delta_r))

def R_from_omega(w):
    return math.exp(m * w**2 * Delta_r / (2 * kB * T))

def tau_mig(w):
    disc = math.sqrt(nu**2 + 4*w**2)
    sp, sm = (-nu + disc)/2, (-nu - disc)/2
    A = r_in * sm / (sm - sp)
    B = r_in * sp / (sp - sm)
    lo, hi = 0.0, 1e-6
    def r(t):
        return A*math.exp(sp*t) + B*math.exp(sm*t)
    while r(hi) < r_out:
        hi *= 2
    for _ in range(80):
        mid = 0.5*(lo+hi)
        lo, hi = (mid, hi) if r(mid) < r_out else (lo, mid)
    return 0.5*(lo+hi)

# R_from_omega(omega_from_R(15)) == 15 to 1e-15
# tau_mig(omega_from_R(15)) == 1.340764e-6
```

Appendix C — nest additivity and work

```
import math, random
N_A, rho = 6.02214076e23, 1000.0
m = 18.01528e-3 / N_A
n = rho / m
N = n

def W(k):
    return N*(k-1)/2 + 26*N/k

# W(100)/ (N*(N-1)/2) == 1/3.36e26

random.seed(0)
k, npart = 100, 1000
vel = [[random.gauss(0,1) for _ in range(3)] for _ in range(npart)]

def PK(vs):
    P = [0.0, 0.0, 0.0]
    K = 0.0
    for v in vs:
        for a in range(3):
            P[a] += m*v[a]
        K += 0.5*m*sum(x*x for x in v)
    return P, K

P, K = PK(vel)
tiles = [PK(vel[j:j+k]) for j in range(0, npart, k)]
Psum = [sum(t[0][a] for t in tiles) for a in range(3)]
Ksum = sum(t[1] for t in tiles)
# |Psum-P|/|P| ~ 2e-15, |Ksum-K|/|K| ~ 1e-15
```